Faiyaj Rahman

U.S. Citizen | 586-455-7796 | faiyajr@umich.edu | linkedin.com/in/faiyajrahman | github.com/faiyajr | faiyajr.dev

Education

University of Michigan – Ann Arbor **Ann Arbor, MI**

B.S.E. in Computer Engineering | GPA: 3.6/4.0 · Dean's List Expected Graduation: May 2029

- **Coursework:** Data Structures & Algorithms, Computer Architecture, Object Oriented Programming, Discrete Math, Computing Systems, Linear Algebra, Multivariable & Vector Calculus

Honors

Jane Street FOCUS | *Selected from 1000+ applicants for quant trading and SWE program* May 2026

Work Experience

Bosch **Farmington Hills, MI**

Software Engineering Intern May 2026 – Aug. 2026

- Deployed a Flask operations platform to AWS and scripted multi-tenant KPIs and analytics scaling to **5000+** devices.
- Designed a Python tool to convert CAN `.asc` logs to `MF4` with DBC channel mapping, saving **20+** engineer-hours monthly.
- Engineered a multi-threaded GUI tool used by **100+** developers to build Python executables, saving **$15K** in licenses.

MRacing FSAE **Ann Arbor, MI**

Software Engineer Aug. 2025 – Present

- Built a C++ Raspberry Pi dashboard decoding **1000+** CAN messages/sec into driver analytics & fault states over **3** pages.
- Collaborated with engineers to integrate vehicle telemetry, CAN diagnostics, and fault checking into the racing dashboard.
- Wrote C firmware for an infrared lap trigger, edge-detecting photosensor voltage drops on beam breaks to timestamp laps.

Hikmah Hackers **Warren, MI**

Co-Founder & Lead Instructor Sep. 2026 – Present

- Co-founded a coding and mentorship program for underserved middle and high school students at a STEM school, teaching programming fundamentals through problem sets & hands-on mini-projects, and giving students tailored career guidance.
- Designing a curriculum spanning CS topics, game/web dev, and AI scripting, featuring hackathons offering cash prizes.

University of Michigan Electric Vehicle Center **Ann Arbor, MI**

Undergraduate Research Software Developer Sep. 2025 – May 2026

- Developed open-source Python UI software for an EV charging module, contributing to **$10K** in net savings for an OEM.
- Designed a Simulink finite-state machine for system-safety logic and charging protocols, translating models into Python.

Saf · Darul Uloom Michigan **Warren, MI**

Software Developer Nov. 2025 – Present

- Launched a school-management beta on Postgres, an Express REST API, and React featuring role-based access control.
- Implemented assignments, grading, and attendance, replacing paper grades for **20+** educators in a **3**-month live pilot.

Projects

LC-2K Pipelined Processor Simulator | *C, Assembly, Make* Nov. 2026 – Present

- Implemented a **5-stage** pipelined processor simulator, with two-state clock updates and separate instruction/data memory.
- Developing forwarding, load-use stalls, and branch squashing; debugging pipeline behavior with targeted assembly tests.

75% ISO Hall-Effect Keyboard PCB | *STM32, KiCad, Solidworks* Dec. 2025 – Jan. 2026

- Designed an STM32 Hall-Effect keyboard PCB, with analog multiplexers, ADC routing, USB, SWD, I^2C OLED header.
- Integrated RGB circuitry, separate analog/digital power distribution, ferrite isolation, decoupling, and ESD protection.

Low-Level Modulator | *Verilog, Assembly, FPGA, Digital Logic* Jan. 2026 – Apr. 2026

- Engineered an **18**-instruction processor in Verilog on a DE2-115 FPGA, driving a synthesizer with **40** assembly modules.
- Designed an integer-math DSP pipeline synthesizing **32-bit** audio from **3** wavetables with **5** modulation modes at **30 FPS**.
- Wrote instructions (arithmetic operations and control logic) for processor execution with FPGA-based audio generation.

Technical Skills

Languages: C, C++, Python, Assembly, Verilog, MATLAB, JavaScript, TypeScript, SQL, Julia, Bash
Tools/Frameworks: STM32, FPGA, CAN/DBC, Raspberry Pi, GPIO, ADC, UART, SPI, I^2C, PCB Design, Altium, KiCad, SolidWorks, Siemens NX, Simulink, Linux, GDB, Valgrind, Make, CMake, GitLab, Docker, Kubernetes, AWS, React, Node/Express, Tailwind, PySide6, Flask, PostgreSQL, Supabase, Claude Code, Codex, Cursor
Practices: Agile/Scrum, Git/GitHub, CI/CD, code review, unit testing, LLM agents